

IMPLEMENTATION PLAN

# Curiosity-Driven Reinforcement Learning for Emergent Tool Use on a Humanoid Robot

Platform: Unitree G1 EDU Humanoid · Simulator: NVIDIA Isaac Lab

|                  |                                                      |
|------------------|------------------------------------------------------|
| Institution      | Purdue University — School of Mechanical Engineering |
| Location         | West Lafayette, IN 47907                             |
| Document         | Full Step-by-Step Implementation Plan                |
| Project Duration | 16 Weeks (4 Phases)                                  |
| Compute Budget   | ~200–500 GPU-hours (NVIDIA A100/H100)                |
| Date             | May 2026                                             |

*This document provides an exhaustive, step-by-step plan for implementing the Curiosity-Driven RL project. Every sub-task, dependency, technical detail, software artifact, hardware procedure, evaluation criterion, risk, and mitigation is captured so that no work item is left ambiguous.*

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# 1. Project Overview & Research Goals

**Project Title:** Curiosity-Driven Reinforcement Learning for Emergent Tool Use on a Humanoid Robot

**Platform:** Unitree G1 EDU (43 DOF, Dex3-1 dexterous hands, NVIDIA Jetson Orin NX onboard, 275 TOPS)

**Simulator:** NVIDIA Isaac Lab (Isaac Sim 4.x) with GPU-parallelised environments

## 1.1 Core Thesis

A full-sized humanoid robot can discover tool-use behaviours *purely through curiosity-driven reinforcement learning*, without demonstrations, pre-programmed heuristics, or dense reward shaping for tool interaction. The robot must independently realise that nearby objects can extend its capabilities — standing on a box to reach an elevated button, hooking a stick to retrieve an occluded item — through intrinsic motivation alone.

## 1.2 Three Core Research Questions

| RQ                | Question                                                                                                                                                                             | Success Criterion                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| RQ1<br>Emergence  | Can a humanoid agent discover tool-use behaviours in sparse-reward environments purely through intrinsic curiosity, without demonstrations or task-specific reward shaping?          | $\geq 60\%$ of seeds independently discover a tool-use solution on $\geq 1$ task |
| RQ2<br>Whole-Body | Does the humanoid morphology enable qualitatively different tool-use strategies vs. fixed-base manipulators — e.g., carrying a tool while walking or repositioning to gain leverage? | Agent exploits bipedal locomotion in $\geq 1$ discovered strategy                |
| RQ3<br>Transfer   | Can curiosity-trained policies transfer to the physical G1 via domain randomisation and asymmetric actor-critic, and do emergent tool-use behaviours survive the reality gap?        | $\geq 30\%$ success rate on physical G1 hardware                                 |

## 1.3 Why This is Novel

- No published paper demonstrates *emergent* tool use on a bipedal humanoid trained from scratch with pure curiosity-driven RL, no demonstrations, and only sparse extrinsic rewards.
- HumanoidBench (RSS 2024) explicitly shows that state-of-the-art RL algorithms — including intrinsic-motivation baselines — *fail* on hard humanoid whole-body manipulation tasks. This project directly attacks that open problem.
- All existing G1 work (ASAP, VideoMimic, DemoHLM, HOMIE) relies on motion-capture data, teleoperation, or at least one demonstration. This work uses **zero demonstrations**.
- The combination of (humanoid bipedal)  $\times$  (curiosity-driven intrinsic reward)  $\times$  (no demos)  $\times$  (sparse extrinsic reward only)  $\times$  (emergent tool use) is unoccupied in the 2023–2026 literature.
- Sim-to-real deployment on the physical Unitree G1 is a high-cost, high-reward differentiator: almost all curiosity papers stop at simulation.

## 1.4 Expected Contributions

- **First demonstration** of curiosity-driven emergent tool use on a full-sized bipedal humanoid.
- **Open-source task suite** (Isaac Lab + G1 URDF) for benchmarking tool-use emergence in humanoid RL.

- **Empirical analysis** of RND vs. DRND vs. RDD vs. ICM vs. QFLEX curiosity mechanisms in high-DOF embodied settings.
- **Sim-to-real pipeline** for multi-phase whole-body behaviours extending the OmniXtreme two-stage architecture to manipulation-centric tasks.
- **Negative-result contribution** if tool use does not emerge: a rigorous characterisation of the 'humanoid hard-exploration' problem — itself publishable as a benchmark akin to Montezuma's Revenge.

## 2. Environment & Hardware Setup (Pre-Phase — Before Week 1)

All items in this section must be completed **before** Phase 1 begins. These are prerequisites, not tracked in the 16-week timeline.

### 2.1 Accounts, Access & Keys

- Obtain Purdue RCAC cluster accounts (Anvil, Gilbreth, or Negishi) for all team members.
- Request GPU allocation:  $\geq 4$  NVIDIA A100 or H100 nodes for parallel Isaac Lab training.
- Verify SSH key-pair setup and two-factor authentication on the cluster.
- Register for an NVIDIA Omniverse / Isaac Lab licence (free for academic use).
- Create a shared GitHub/GitLab private repository with main, dev, and experiment branches.
- Set up Weights & Biases (wandb) project for experiment tracking — create API keys for all members.
- Set up a shared network drive or S3 bucket for checkpoint storage ( $\geq 1$  TB recommended).

### 2.2 Local Development Machines

- Each workstation should have  $\geq 1$  NVIDIA GPU (RTX 3090 or better) for local debugging.
- Install Ubuntu 22.04 LTS (required by Isaac Lab and ROS 2 Humble).
- Install NVIDIA driver 525+, CUDA 12.1+, cuDNN 8.9+.
- Install Docker 24+ with NVIDIA Container Toolkit for reproducible environments.
- Clone NVIDIA Isaac Lab repository and run the built-in sanity-check scene.
- Clone unitree\_rl\_gym (official Unitree RL repository) and run the G1 standing task.
- Install MuJoCo 3.x locally for sim-to-sim cross-validation (separate conda env).
- Install TensorRT for onboard inference testing (match Jetson Orin NX version).

### 2.3 Hardware Inventory Check

| Item                            | Required Spec                                                       | Status Check                                                    |
|---------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|
| Unitree G1 EDU robot            | 43 DOF, Dex3-1 hands, Jetson Orin NX                                | Confirm availability; if to-procure, initiate order immediately |
| Safety gantry / overhead tether | Load-rated for $\geq 70$ kg robot + dynamic loads                   | Inspect mount points; load-test before any real-robot run       |
| Tool objects                    | Wooden blocks 30×30×30 cm (1–3 kg); PVC rods 80–120 cm; foam wedges | Fabricate or procure; label with mass/friction properties       |
| Ethernet switch + cables        | 1 Gbps for unitree_sdk2_python communication                        | Confirm low-latency ( $< 1$ ms) on test bench                   |
| E-stop remote                   | Wireless, $< 50$ ms latency                                         | Test before any untethered robot motion                         |
| RGB-D camera (external)         | Optional — for environment monitoring, not policy input             | Check mount positions relative to task scene                    |

### 2.4 Software Dependency Versions (Pin These)

| Package             | Version                         | Notes                                     |
|---------------------|---------------------------------|-------------------------------------------|
| Isaac Lab           | 4.x (Isaac Sim 4.x backend)     | GPU-parallel RL environments              |
| unitree_rl_gym      | Latest main (2025)              | G1 URDF + PPO boilerplate                 |
| rsl_rl              | 1.0.x                           | PPO implementation used by unitree_rl_gym |
| MuJoCo              | 3.x                             | Sim-to-sim cross-validation               |
| TensorRT            | 8.6+ (match Jetson Orin NX SDK) | Onboard inference optimisation            |
| unitree_sdk2_python | Latest                          | Hardware communication layer              |
| ROS 2 Humble        | Humble Hawksbill                | Hardware state/command bridge             |
| PyTorch             | 2.1+                            | Neural network training                   |
| wandb               | Latest                          | Experiment tracking                       |

## 3. Phase 1 — Environment Design in Isaac Lab (Weeks 1–3)

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Goal: Deliver a fully validated Isaac Lab environment suite containing all 5 tasks, with the G1 URDF integrated, procedural object randomisation implemented, and training infrastructure tested on the cluster ( $\geq 1$  GPU,  $\geq 256$  parallel envs confirmed).

### 3.1 Software Installation & Infrastructure (Week 1, Days 1–3)

#### Install Isaac Lab on cluster

- Pull the official NVIDIA Isaac Lab Docker image onto the cluster.
- Configure SLURM job scripts for multi-GPU Isaac Lab training (4096 envs target).
- Run the provided Isaac Lab sanity-check scene; confirm GPU utilisation  $\geq 80\%$ .
- Install conda environment with all Python dependencies (torch, rsl\_rl, wandb, numpy).
- Install unitree\_rl\_gym and run G1 standing task — confirm the G1 URDF loads without errors.

#### Set up version control & CI

- Create project repository structure: src/envs/, src/agents/, src/distill/, src/sim2real/, configs/, scripts/.
- Add pre-commit hooks: black formatter, flake8 linter, unit-test runner.
- Set up GitHub Actions (or cluster-native CI) to run unit tests on every push.
- Tag commit v0.0.1 — baseline: Isaac Lab confirmed running with G1 URDF.

#### Set up experiment tracking

- Create wandb project 'CDL-HumanoidToolUse'.
- Define standard logging keys: episode\_reward, intrinsic\_reward, extrinsic\_reward, success\_rate, tool\_contact\_count, alpha\_annealing.
- Write a lightweight experiment launcher script (launch.py) that accepts a YAML config and logs to wandb.
- Test wandb sweep agent on dummy environment.

### 3.2 G1 URDF Integration & Validation (Week 1, Days 4–5)

- Import G1 URDF (43 DOF) into Isaac Lab using the unitree\_rl\_gym URDF path.
- Verify all 43 joints are correctly parsed: hip, knee, ankle (6 DOF per leg), waist (3 DOF), shoulder, elbow, wrist (7 DOF per arm), Dex3-1 finger joints.
- Confirm Dex3-1 hand joint limits, friction, and gear ratio are correctly set from the hardware datasheet.
- Implement a PD controller test: command all joints to zero position; verify no physics instabilities.
- Measure G1 reach envelope in simulation (should be  $\approx 1.6$  m max reach): confirm it matches the proposal spec.
- Confirm IMU sensor (orientation, angular velocity) is correctly attached to the pelvis link.
- Confirm wrist force-torque sensor is correctly attached to both wrists.
- Confirm the head camera (64×64 egocentric depth image at 50 Hz) is correctly placed and outputs valid depth.
- Run 100-step rollout with random actions; confirm no NaN in observations or actions.
- Document joint indices, observation vector layout, and action space bounds in a dedicated doc-string / README.

### 3.3 Task Environment Implementation — All 5 Tasks (Weeks 1–2)

Each task is implemented as a separate Isaac Lab environment class inheriting from a shared ToolUseBase environment. The base class handles G1 spawning, physics settings, observation computation, reward computation, and reset logic.

| Task                  | Goal Description                                                   | Latent Solution                                                                 | Difficulty | Key Implementation Details                                                                        |
|-----------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------|
| 1. Distant Target     | Touch a target mounted beyond arm's reach across a 0.6 m gap       | Pick up a long stick (80–120 cm PVC rod) to bridge the gap                      | Easy       | Stick must be placeable anywhere; target is a force-sensitive pad. Use as curriculum Stage 1.     |
| 2. Elevated Button    | Press a button mounted 2.0 m high (G1 max reach $\approx 1.6$ m)   | Push a box underneath, climb onto it, then press button                         | Medium     | Box must support G1 weight ( $\geq 80$ kg). Implement contact-force check on box top surface.     |
| 3. Occluded Retrieval | Retrieve an object placed behind a narrow barrier/gap (15 cm slot) | Use stick/rod to push or hook the object out                                    | Medium     | Barrier is rigid. Object must move $\geq 0.5$ m to count as retrieved. Track object displacement. |
| 4. Weight Lever       | Lift a 10 kg object beyond G1's direct payload capacity            | Use a plank as a lever with a fulcrum object                                    | Hard       | Implement torque/contact sensing on plank. Fulcrum object must be correctly placed by agent.      |
| 5. Composite          | Navigate past an obstacle, then reach an elevated goal             | Sequence: carry box $\rightarrow$ place $\rightarrow$ climb $\rightarrow$ reach | Hard       | Combine Elevated Button + locomotion. Long episode horizon (3000 steps). Add to curriculum last.  |

#### For each task, the implementation checklist is:

- Write environment class (e.g., ElevatedButtonEnv) with `__init__`, `reset()`, `step()`, `compute_observations()`, `compute_reward()`.
- Spawn all objects (goal, tools, distractors) using Isaac Lab's asset API.
- Implement procedural randomisation of object positions, orientations, masses, and friction (see Section 3.4).
- Implement binary extrinsic reward: +1 only on goal completion, 0 otherwise; time penalty  $-0.001/\text{step}$ .
- Implement success detection logic (force threshold, object displacement, button press signal).
- Write unit test: reset 100 times, confirm no interpenetration or NaN observations.
- Run 10-minute training run with random policy and confirm  $\leq 0.1\%$  accidental success rate (task must be non-trivially hard).
- Visually inspect 5 rollouts in the Isaac Lab viewer to confirm physics looks correct.
- Log task-specific diagnostic: `tool_contact_events`, `goal_proximity_over_time`.

### 3.4 Object Randomisation & Reward Specification (Week 2)

Randomisation prevents the policy from memorising fixed object configurations and forces learning a generalised tool-use strategy.

| Property             | Range                 | Distribution | Purpose                                |
|----------------------|-----------------------|--------------|----------------------------------------|
| Object position (XY) | $\pm 30\%$ of nominal | Uniform      | Prevent memorisation of exact location |



| Property              | Range                                | Distribution  | Purpose                                  |
|-----------------------|--------------------------------------|---------------|------------------------------------------|
| Object orientation    | Full SO(2) yaw randomisation         | Uniform       | Force orientation-agnostic grasping      |
| Object mass           | $\pm 30\%$ of nominal                | Uniform       | Generalise to real object mass variance  |
| Object friction       | $\pm 50\%$ of nominal                | Uniform       | Sim-to-real friction gap                 |
| Object colour/texture | Random from 8-colour palette         | Categorical   | Prevent visual cue overfitting           |
| Goal position         | $\pm 10\%$ of nominal (height fixed) | Uniform       | Slight variance for robustness           |
| Distractor objects    | 1–3 random objects in scene          | Random choice | Prevent tool identification by exclusion |

### Extrinsic Reward Design (Binary Sparse):

- $r_{\text{extrinsic}} = +1.0$  upon goal completion (task-specific success condition), 0 otherwise.
- $r_{\text{time}} = -0.001$  per timestep (encourages efficiency; does NOT guide tool-use strategy).
- No intermediate reward shaping for picking up tools, approaching objects, or making contact.
- Episode terminates on success, on robot fall (pelvis height  $< 0.5$  m), or at 2,000 steps.
- For the Composite task: episode terminates at 3,000 steps.

### 3.5 Phase 1 Deliverables & Exit Criteria

- All 5 Isaac Lab environments implemented, unit-tested, and committed to the repository.
- G1 URDF validated: all 43 DOF correct, sensors confirmed, PD controller stable.
- Training infrastructure tested: SLURM job script runs 4,096 parallel envs on  $\geq 4$  A100 GPUs.
- wandb project live: at least one dummy-agent training run logged with all required metrics.
- Random-agent baseline success rates measured and documented ( $< 0.1\%$  for hard tasks).
- Phase 1 review meeting: demo 5 Isaac Lab environments in real-time viewer to advisor.

**Exit criterion:** Phase 2 begins only when all 5 environments pass unit tests AND the cluster can sustain  $\geq 4,096$  parallel Isaac Lab environments for  $\geq 1$  hour without crash.

## 4. Phase 2 — Curiosity-Driven RL Training (Weeks 4–9)

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Goal: Train curiosity-driven policies across the full task suite. Achieve at least one clear instance of emergent tool-use behaviour ( $\geq 20\%$  success rate on Distant Target task with RND/DRND). Complete all ablation runs and hyperparameter sweeps.

### 4.1 Algorithm Implementation — Curiosity Module (Week 4)

We implement three curiosity mechanisms to enable the ablation study. All are implemented as modular PyTorch nn.Module classes that plug into the training loop.

#### 4.1.1 RND — Random Network Distillation (Primary, Burda et al. 2019)

- Implement fixed random target network  $f_{\text{target}}$ :  $\text{obs} \rightarrow \mathbb{R}^d$  ( $d=512$  for image obs,  $d=64$  for state-vector obs).
- Implement trainable predictor network  $f_{\text{pred}}$ :  $\text{obs} \rightarrow \mathbb{R}^d$  (same architecture as target).
- Intrinsic reward:  $r_{\text{int}} = \|f_{\text{pred}}(o) - f_{\text{target}}(o)\|^2$  per timestep.
- Predictor network is trained with MSE loss; target network is frozen.
- Apply running mean/variance normalisation to  $r_{\text{int}}$  to prevent reward scale explosion.
- Implement separate value head for intrinsic returns (do NOT mix intrinsic and extrinsic returns in a single head — use two separate critics).
- Unit test: confirm  $r_{\text{int}}$  decreases monotonically as agent revisits the same state.

#### 4.1.2 DRND — Distributional RND (Yang et al., ICML 2024)

- Implement an ensemble of  $N=10$  random target networks (each with different random initialisations).
- Compute intrinsic bonus as:  $r_{\text{int}} = \text{Var}(f_{\text{target}_i}(o))$  — high variance = novel state.
- This 'bonus inconsistency' fix addresses vanilla RND's over-rewarding of already-explored states.
- Implement same two-head critic structure as RND.
- Add DRND as a command-line flag: `--curiosity=drnd`.
- Baseline comparison: confirm DRND outperforms RND on Distant Target task within 100M steps.

#### 4.1.3 RDD — Random Distribution Distillation (Fang et al. 2025, arXiv:2505.11044)

- The target network outputs a *distribution*  $N(\mu_{\theta}(s), \sigma^2)$  rather than a fixed vector.
- Sample  $f_{\text{tar}}(s) \sim N(\mu_{\theta}(s), \sigma^2 \cdot I)$  at each step.
- Predictor minimises MSE to the sampled target output.
- Intrinsic bonus:  $b(s) = (1/d) \|f_{\theta}(s) - \mu_{\theta}(s)\|^2$ .
- This unifies count-based and prediction-error exploration: the pseudo-count term  $(\sigma^2/n)$  ensures proper exploration decay; the discrepancy term captures predictor convergence.
- RDD is theoretically stronger than both RND and DRND and should be the primary comparator against vanilla RND.
- Set  $\sigma=1.0$  (fixed), output dim  $d=512$  (image) or  $d=64$  (state-vector) per the RDD ablation results.
- Add as `--curiosity=rdd` flag.

#### 4.1.4 ICM — Intrinsic Curiosity Module (Pathak et al. 2017) — Ablation Baseline

- Implement a forward model: predict next observation embedding from current observation + action.
- Intrinsic reward = forward model prediction error.

- Also implement the inverse model (action prediction) as regularisation.
- Add as --curiosity=icm flag.
- ICM is included for the ablation study (RQ1) — not expected to be the best performer.

## 4.2 Asymmetric Actor-Critic Architecture (Week 4)

The key architectural insight: the **Actor** (deployed policy) sees only what is available on the physical robot, while the **Critic** (training only) receives privileged simulator information.

| Component                            | Inputs                                                                                                                                                                                                                                                | Architecture                                                                                                                             | Output                                              | Deployed ?                    |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------|
| Actor (deployed policy)              | • Proprioception: joint positions (43), velocities (43), torques (43) • IMU: orientation (4-quat), angular velocity (3) • Wrist force-torque: left (6) + right (6) • Depth image: 64×64 egocentric (head camera) Total: ≈165 + 4096 (image flattened) | MLP: [512, 256, 128] + CNN encoder for depth image (4 conv layers → 256-dim latent). Concatenate proprioceptive features + image latent. | Target joint positions at 50 Hz (43-dim continuous) | YES — runs on Jetson Orin NX  |
| Critic (training only)               | All actor inputs PLUS privileged: • Full object poses (position + quaternion for all objects) • Contact states (which links are in contact) • Goal position (exact 3D coordinate) • Terrain mesh info                                                 | MLP: [512, 256, 128]. Receives concatenated actor obs + privileged info.                                                                 | Value estimate V(s) for GAE advantage computation   | NO — discarded after training |
| RND/DRND/RD D Module                 | Actor observation vector (no privileged info — keeps sim-to-real compatible)                                                                                                                                                                          | Target net: [256, 256] → d-dim. Predictor net: [256, 256] → d-dim.                                                                       | Intrinsic reward scalar r_int per step              | NO — training only            |
| Intrinsic Critic (second value head) | Actor observations + intrinsic reward stream                                                                                                                                                                                                          | Shares backbone with extrinsic critic; separate output head.                                                                             | V_int(s) — separate from V_ext(s)                   | NO                            |

## 4.3 Observation & Action Space Specification (Week 4)

**Action Space:** Continuous, 43-dimensional. Each dimension is a target joint position (PD-controlled at 50 Hz). Actions are clipped to joint limits. Apply action smoothing (exponential moving average with  $\alpha=0.8$ ) to prevent jerky motion.

**Observation Normalisation:** Apply running mean/variance normalisation independently to each observation channel. This is critical for stable training with mixed-scale inputs (e.g., joint angles in radians vs. force-torque in Newtons).

**Episode Horizon:** 2,000 steps (40 seconds at 50 Hz) for tasks 1–4. 3,000 steps (60 seconds) for the Composite task.

## 4.4 Reward Annealing & Curriculum Staging (Weeks 4–6)

**Total reward:**  $r_{\text{total}} = r_{\text{extrinsic}} + \alpha \cdot r_{\text{intrinsic}}$

**$\alpha$  annealing schedule:**  $\alpha$  is linearly annealed from 1.0 → 0.1 over the full training run. This shifts the agent from exploration to exploitation as it discovers successful strategies. Monitor  $\alpha$  on wandb.

**Curriculum Staging (unlock sequence):**

| Stage | Task Unlocked               | Unlock Condition                   | Approx. Steps |
|-------|-----------------------------|------------------------------------|---------------|
| 1     | Distant Target (Easy)       | Start of training                  | 0             |
| 2     | Elevated Button (Medium)    | >20% success on Distant Target     | ~50–100M      |
| 3     | Occluded Retrieval (Medium) | >20% success on Elevated Button    | ~100–200M     |
| 4     | Weight Lever (Hard)         | >20% success on Occluded Retrieval | ~200–350M     |
| 5     | Composite (Hard)            | >20% success on Elevated Button    | ~200–500M     |

*Note: Stages 4 and 5 may run in parallel once Stage 3 unlocks. All tasks share the same policy; curriculum adjusts the distribution of environment instances allocated to each task.*

## 4.5 Training Infrastructure & Cluster Setup (Week 4–5)

- Write SLURM job script: 4 A100 nodes, 4 GPUs each, 48-hour wall time, auto-requeue on timeout.
- Configure Isaac Lab for 4,096 parallel environments distributed across GPUs (1,024 per GPU).
- Implement checkpoint saving every 10M steps and on every new peak success rate.
- Implement automatic experiment resumption from checkpoint if job is killed (SLURM preemption).
- Set episode horizon to 2,000 steps; batch size = 4,096 envs × 16 steps = 65,536 transitions per update.
- PPO update: 10 mini-epochs per rollout, mini-batch size 8,192.
- Estimated training throughput: ~10M steps/hour on 4×A100; full 500M step run ≈50 hours.
- Run a 10M-step smoke test on the cluster before launching full runs.
- Monitor GPU utilisation, memory, and step/sec in real-time via wandb system metrics.

## 4.6 Hyperparameter Sweeps & Ablation Runs (Weeks 5–8)

| Hyperparameter                      | Search Range       | Default | Search Method |
|-------------------------------------|--------------------|---------|---------------|
| Learning rate (actor)               | [1e-4, 3e-4, 1e-3] | 3e-4    | Grid          |
| Learning rate (critic)              | [3e-4, 1e-3]       | 1e-3    | Grid          |
| GAE $\lambda$                       | [0.90, 0.95, 0.98] | 0.95    | Grid          |
| PPO clip $\epsilon$                 | [0.1, 0.2, 0.3]    | 0.2     | Grid          |
| Discount $\gamma_{\text{ext}}$      | [0.99, 0.999]      | 0.999   | Grid          |
| Discount $\gamma_{\text{int}}$      | [0.99, 0.999]      | 0.99    | Grid          |
| $\alpha$ initial (curiosity weight) | [0.5, 1.0, 2.0]    | 1.0     | Grid          |
| $\alpha$ final (after annealing)    | [0.05, 0.1, 0.2]   | 0.1     | Grid          |
| RND/DRND/RDD embedding dim d        | [64, 256, 512]     | 512     | Grid          |
| DRND ensemble size N                | [5, 10, 20]        | 10      | Grid          |
| Action smoothing $\alpha$           | [0.6, 0.8, 0.9]    | 0.8     | Grid          |

*Run sweeps using wandb Sweeps (Bayesian optimisation after initial grid). Run each configuration for 100M steps with 3 seeds. Select best configuration for full 500M-step training runs.*

**Ablation study design (4 ablations, each run for 200M steps, 5 seeds):**

| Ablation                | Variant A                        | Variant B                             | Research Question                                       |
|-------------------------|----------------------------------|---------------------------------------|---------------------------------------------------------|
| A1: Curiosity mechanism | RND vs. DRND vs. RDD vs. ICM     | No curiosity (PPO only)               | Which curiosity method works best on high-DOF humanoid? |
| A2: Critic design       | Privileged asymmetric critic     | Symmetric actor-critic (no privilege) | Does privileged critic accelerate learning?             |
| A3: Visual input        | Ego depth image + proprioception | Proprioception only (no depth)        | Does visual input accelerate tool discovery?            |
| A4: Curriculum          | Staged curriculum (easy→hard)    | All tasks simultaneously              | Does curriculum staging improve emergence rate?         |

## 4.7 Emergence Analysis & Logging (Weeks 6–9)

- Define 'tool-use emergence event': first episode where the agent (a) makes contact with a tool object AND (b) successfully completes the task within the same episode. Log the training step at which this first occurs.
- Log tool\_contact\_count (number of timesteps per episode where a tool is grasped) over training.
- Log tool\_use\_rate = fraction of successful episodes that involved tool contact.
- Every 50M steps, run 100 evaluation episodes (no exploration noise) and log success\_rate per task.
- At end of training, run 10 independent seeds and compute Tool Use Emergence Rate (% of seeds that discovered tool use).
- Save video rollouts every 50M steps using Isaac Lab's built-in recorder.
- Implement trajectory clustering: use k-means (k=5) on 50-dim trajectory embeddings to identify distinct tool-use strategies per task. Target:  $\geq 2$  distinct strategies per task.
- Run HumanoidBench intrinsic-motivation baselines for comparison if possible (confirms our method against the existing benchmark).

## 4.8 Phase 2 Deliverables & Exit Criteria

- RND, DRND, RDD, and ICM modules fully implemented, tested, and committed.
- Full 500M-step training run completed on  $\geq 1$  task with  $\geq 1$  curiosity method.
- Tool use emergence observed in at least one task ( $\geq 20\%$  success rate on Distant Target as minimum bar).
- All 4 ablation variants run for 200M steps with 5 seeds each.
- Hyperparameter sweep completed; best config identified.
- wandb dashboard showing all training curves, emergence events, and ablation comparisons.
- Video recordings of at least 3 clear emergent tool-use episodes saved.
- Phase 2 review meeting: present learning curves and best-performing episodes to advisor.

**Exit criterion:** Phase 3 begins when  $\geq 1$  task has achieved  $>20\%$  success rate with a curiosity-driven policy AND  $\geq 3$  clean tool-use rollouts are recorded.

## 5. Phase 3 — Policy Distillation & Robustification (Weeks 10–11)

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Goal: Convert the noisy curiosity-driven policy into a clean, deployable policy via DAgger-based behaviour cloning, then refine it with residual RL under hardware-realistic actuator constraints. Cross-validate in MuJoCo to confirm behaviours are not simulator-specific artifacts.

### 5.1 Phase A — Behaviour Cloning via DAgger (Week 10, Days 1–4)

The curiosity-driven policy uses exploration noise (RND overhead, random perturbations) that make it unsuitable for hardware deployment. DAgger distills its *behaviour* into a clean student policy without exploration overhead.

- Collect demonstration dataset  $D$  from the best-performing curiosity policy: run 10,000 episodes with the trained actor (deterministic, no exploration noise); save (observation, action) pairs.
- Filter dataset: keep only episodes with tool-use success (final  $r_{\text{extrinsic}} = +1$ ) to ensure the student learns successful strategies, not failed explorations.
- Train a student policy (same architecture as actor, NO RND module) via behaviour cloning: minimise  $\|\pi_{\text{student}}(o) - \pi_{\text{teacher}}(o)\|^2$  using Adam,  $\text{lr}=1\text{e-}4$ , for 100 epochs.
- After each epoch, collect new rollouts with the student policy and add to dataset  $D$  (the DAgger loop: student-generated states prevent distribution shift).
- Run DAgger for 20 iterations (each iteration: 500 episodes rollout + 50 epochs BC training).
- Evaluate student policy every 5 DAgger iterations: confirm success rate tracks teacher within 10%.
- Confirm the student policy has NO RND module and runs inference in  $<10$  ms (required for 50 Hz control).
- Save the distilled student policy checkpoint as `student_policy_v1.pt`.

### 5.2 Phase B — Residual RL Refinement (Week 10, Days 5 – Week 11, Day 3)

The distilled base policy may have sim-to-real gaps due to idealised actuator models. Residual RL fine-tunes a small MLP on top of the frozen base policy under realistic actuator constraints.

- **Freeze base policy:** `student_policy_v1.pt` weights are frozen — no gradient flows through them.
- **Add residual MLP:** a lightweight 2-layer MLP [128, 64] takes the base policy's action + current observation and outputs a residual correction  $\delta a$ . Final action =  $\pi_{\text{base}}(o) + \delta a$ .
- **Actuator constraints applied during residual training:**
  - Torque-speed curves: joint torque limits from the G1 hardware spec (varies per joint).
  - Overcurrent protection: clip torque if sustained for  $>0.5$  s above thermal limit.
  - Joint position/velocity limits: hard clipped to G1 hardware limits.
  - Communication latency: simulate 0–20 ms random action delay.
  - Motor response curves: apply first-order lag filter ( $\tau=0.02$  s) to action execution.
- **Domain randomisation for residual training:**
  - Mass:  $\pm 20\%$  of nominal for all links.
  - Friction (ground + objects):  $\pm 40\%$  of nominal.
  - Joint damping:  $\pm 30\%$  of nominal.
  - Observation noise: Gaussian noise  $\sigma=0.01$  on all proprioceptive channels.
  - Gravity direction:  $\pm 3^\circ$  tilt from vertical.
- Train residual MLP with PPO for 50M steps (much shorter than curiosity training — base provides strong prior).

- Evaluate: success rate of (base + residual) vs. base alone; confirm improvement  $\geq 10\%$ .
- Save final checkpoint: `deployable_policy_v1.pt` = frozen base + trained residual.

### 5.3 Sim-to-Sim Cross-Validation in MuJoCo (Week 11, Days 4–5)

Deploy the trained policy in MuJoCo 3.x (independent of Isaac Lab) to verify that behaviours are not simulator-specific artifacts. If the policy fails completely in MuJoCo, it signals a physics modelling gap that must be resolved before hardware deployment.

- Convert G1 URDF to MuJoCo XML format using the `mujoco-compile` tool.
- Implement the same 5 task environments in MuJoCo (simplified versions acceptable — same object geometry, same reward structure).
- Load `deployable_policy_v1.pt` and run 100 evaluation episodes in MuJoCo.
- Target:  $\geq 50\%$  of the Isaac Lab success rate achieved in MuJoCo.
- If success rate drops  $> 50\%$  in MuJoCo, investigate the specific physics discrepancy (contact model, friction, actuator model) and apply targeted domain randomisation fixes.
- Document MuJoCo success rate in the paper as evidence against simulator overfitting.

### 5.4 Phase 3 Deliverables & Exit Criteria

- DAgger distillation complete: clean student policy with no RND overhead, inference  $< 10$  ms.
- Residual RL training complete: `deployable_policy_v1.pt` saved.
- MuJoCo cross-validation complete:  $\geq 50\%$  of Isaac Lab success rate achieved.
- Policy inference latency confirmed  $< 10$  ms on a local GPU (TensorRT export is Phase 4).
- Phase 3 review: demonstrate clean policy rollout (no exploration jitter) in Isaac Lab viewer.

**Exit criterion:** Phase 4 begins when `deployable_policy_v1.pt` achieves  $\geq 50\%$  success rate in MuJoCo AND inference is confirmed  $< 10$  ms on CPU-equivalent hardware.

## 6. Phase 4 — Sim-to-Real Transfer (Weeks 12–14)

Goal: Deploy the trained policy on the physical Unitree G1 EDU robot. Conduct a systematic real-world evaluation protocol. Document successes and failures. Achieve  $\geq 30\%$  success rate on the physical robot on  $\geq 1$  task.

### 6.1 Domain Randomisation Ranges for Sim-to-Real (Already Applied in Phase 3)

| Parameter                      | Sim Range                            | Purpose                          |
|--------------------------------|--------------------------------------|----------------------------------|
| Link masses                    | $\pm 20\%$ of CAD spec               | Compensate for payload variation |
| Ground friction ( $\mu$ )      | $\pm 40\%$ (0.3–1.0)                 | Different floor surfaces         |
| Joint damping                  | $\pm 30\%$ of nominal                | Actuator wear variation          |
| Observation noise ( $\sigma$ ) | Gaussian, $\sigma=0.01$ all channels | Encoder/IMU noise                |
| Action delay                   | 0–20 ms random                       | Communication latency            |
| Gravity direction              | $\pm 3^\circ$ from vertical          | Uneven floor, calibration error  |
| Object mass (real tools)       | $\pm 30\%$ of labelled mass          | Manufacturing tolerance          |
| Object friction (real tools)   | $\pm 50\%$ of nominal                | Surface variation                |

### 6.2 Safety Protocol (MUST be followed before ANY robot power-on)

■ ■ All real-world tests carry risk of hardware damage and physical injury. The following protocol is MANDATORY and must be reviewed by the faculty advisor before any hardware session.

- **Pre-session checklist (complete before every session):**

- Inspect all robot joints for mechanical wear or unusual resistance.
- Check Dex3-1 finger integrity — no bent linkages or loose screws.
- Verify the safety gantry is load-rated and tether is attached to the robot's harness point.
- Confirm the emergency stop (E-stop) remote is charged and within reach of all team members.
- Confirm the lab is clear of non-essential personnel.
- Power on robot in damping mode (joints limp) first; confirm no error LEDs.
- Run factory self-test via `unitree_sdk2_python`; confirm all actuator IDs respond.

- **Behavioural kill-switch implementation:**

- Implement a software kill-switch that monitors joint torque in real-time (100 Hz).
- If any joint exceeds 90% of its maximum rated torque for  $>100$  ms, immediately send zero-torque command.
- If centre-of-mass deviation exceeds  $\pm 0.3$  m from the tether attachment point, send zero-torque command.
- If any foot contact force exceeds  $3\times$  body weight, send zero-torque command.
- The E-stop remote cuts power to all motors immediately (hardware-level, not software).

- **Graduated deployment progression:**

- Stage 1 (Week 12): Static tool interaction — robot is standing still, tethered. Test arm reaching and grasping motions only. No locomotion.
- Stage 2 (Week 13): Slow-motion full task — run policy at 25% speed (override control loop to command  $0.25\times$  the policy's action magnitudes). Observe for instabilities.
- Stage 3 (Week 14): Full speed — run policy at 100% if Stage 2 shows no instabilities across 20 consecutive episodes.



- At no stage should untethered bipedal locomotion be attempted unless the faculty advisor approves.

## 6.3 Hardware Deployment Procedure (Weeks 12–14)

- **TensorRT export (do this before Week 12):**

- Export deployable\_policy\_v1.pt to TensorRT engine for the Jetson Orin NX.
- Target inference latency:  $\leq 10$  ms (50 Hz control requires action computation in  $< 20$  ms total).
- Validate TensorRT output matches PyTorch output within  $1e-4$  absolute error on 1000 test inputs.
- Flash the TensorRT engine to the Jetson Orin NX onboard the G1.

- **Communication setup:**

- Connect development laptop to G1 via Ethernet (1 Gbps).
- Run unitree\_sdk2\_python control loop: subscribe to joint state topic, publish joint position commands.
- Confirm round-trip latency  $< 5$  ms on the Ethernet link.
- Implement ROS 2 Humble bridge node: `/joint_states`  $\rightarrow$  policy  $\rightarrow$  `/joint_commands`.
- Implement real-time logging: record all observations and actions to a local bag file for post-hoc analysis.

- **Observation computation on Jetson:**

- Proprioceptive observations computed directly from unitree\_sdk2\_python joint state.
- IMU data: orientation from built-in IMU (quaternion), angular velocity from built-in gyroscope.
- Wrist force-torque: read from Dex3-1 wrist sensors via SDK.
- Depth image: capture from head camera at 50 Hz; resize to  $64 \times 64$ ; apply same normalisation as training.
- Concatenate all observations and feed to TensorRT engine.

- **Task scene setup (physical lab):**

- Arrange the physical task scene matching the simulation layout (measured to  $\pm 2$  cm).
- Label all tool objects with their mass and confirm it matches the simulation  $\pm 30\%$  range.
- Measure friction coefficients of physical objects with a tilt-test; confirm within  $\pm 50\%$  range.
- Place safety padding around all task objects to prevent damage on robot fall.
- Set up an external overhead camera for video documentation (not used by the policy).

## 6.4 Real-World Evaluation Protocol (Weeks 13–14)

Conduct a systematic evaluation protocol matching the simulation evaluation to enable direct comparison.

- Run 50 episodes per task on the physical G1 (prioritise Distant Target and Elevated Button tasks).
- For each episode: reset the task scene to a randomised configuration (within  $\pm 30\%$  position,  $\pm 50\%$  friction).
- Record: success (binary), episode length, tool contact events (from force-torque sensor), any safety-kill activations.
- Compute real-world success rate and compare to simulation success rate.
- Record high-quality video of every successful trial for the paper's supplementary video.
- If success rate  $< 10\%$  on the physical robot after 50 episodes: invoke risk mitigation protocol (see Section 10).
- Conduct qualitative analysis: compare the robot's tool-use strategy in sim vs. real — does the same strategy (e.g., approaching from the same angle) emerge?

## 6.5 Phase 4 Deliverables & Exit Criteria

- TensorRT policy running on Jetson Orin NX at  $\leq 10$  ms latency.
- 50 real-world evaluation episodes completed per task (at minimum Distant Target + Elevated Button).
- Video documentation:  $\geq 5$  successful real-world tool-use episodes recorded in high quality.
- Real-world success rate measured and documented: target  $\geq 30\%$  on  $\geq 1$  task.

- Safety log: record all E-stop activations and kill-switch triggers.
- Phase 4 review: live demo or video demo of real-robot tool use to advisor.

**Exit criterion:** Phase 5 begins when  $\geq 1$  task shows  $\geq 10\%$  real-world success rate (relaxed from 30% target to allow paper writing to begin in parallel).

## 7. Phase 5 — Paper Writing & Code Release (Weeks 15–16)

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### Week 15 — Draft:

- Write Abstract (confirm final claims match achieved results).
- Write Introduction (motivation, gap statement, contributions — use HumanoidBench as headline evidence).
- Write Related Work (5 subsections: curiosity RL methods; emergent tool use; humanoid whole-body RL; sim-to-real; benchmarks — cite all 26 papers from the literature review).
- Write Method section (4.1 env design, 4.2 curiosity module, 4.3 architecture, 4.4 training details).
- Generate all figures: pipeline diagram, task suite figure, learning curves, ablation plots, trajectory clustering visualisations, real-robot photo/video stills.
- Write Experiments section: environment descriptions, training setup, all quantitative results tables.
- Internal draft review: all co-authors read and comment.

### Week 16 — Finalise & Release:

- Revise draft based on internal review.
- Write Conclusion and Limitations sections.
- Prepare supplementary: full hyperparameter table, additional ablation plots, extended video.
- Produce supplementary video: 3–5 minutes showing simulation training, emerged behaviours, and real-robot deployment.
- Code release preparation: clean up codebase, write README with installation and training instructions, add MIT or Apache 2.0 license, upload to GitHub.
- Submit to target venue (ICRA / IROS / CoRL — select based on timeline).
- Archive checkpoints and datasets on Purdue shared storage for reproducibility.

### Target Venues (in order of preference):

- CoRL 2026 (Conference on Robot Learning) — submission deadline typically July 2026. Best fit: manipulation + RL + sim-to-real.
- IROS 2026 (IEEE/RSJ Intelligent Robots and Systems) — submission typically March 2026. High impact for hardware demonstration papers.
- ICRA 2027 — submission typically September 2026. Largest robotics venue.
- NeurIPS 2026 (Robotics workshop) — for initial rapid dissemination if full paper not ready.

## 8. Evaluation Plan & Success Metrics

| Metric                       | Definition                                                                              | Target                     | Measurement Method                                                                     | Phase   |
|------------------------------|-----------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------|---------|
| Tool Use Emergence Rate      | % of seeds (out of 10) that independently discover a tool-use solution on $\geq 1$ task | $\geq 60\%$                | 10 independent seeds; each run 500M steps; classify as 'emerged' if tool_use_rate > 5% | Phase 2 |
| Task Success Rate (sim)      | % of episodes completed successfully after policy convergence (200 eval episodes)       | $\geq 50\%$ (sim)          | Run 200 eval episodes with no exploration noise; count successes                       | Phase 2 |
| Exploration Efficiency       | Number of training steps to first tool-use discovery event                              | $< 500\text{M steps}$      | Log first episode with tool_contact AND success; record training step                  | Phase 2 |
| Behavioural Diversity        | Number of distinct tool-use strategies per task (k-means on trajectory embeddings)      | $\geq 2$ per task          | k-means (k=5) on 50-dim trajectory embeddings from 1000 successful episodes            | Phase 2 |
| Sim-to-Real Transfer Rate    | Success rate of tool-use behaviours on physical G1 hardware                             | $\geq 30\%$                | 50 real-world evaluation episodes per task                                             | Phase 4 |
| Ablation Delta               | Success rate improvement of best curiosity method vs. no-curiosity PPO baseline         | $\geq 3\times$ improvement | 200M step runs, 5 seeds each; compare best curiosity vs. PPO-only                      | Phase 2 |
| Inference Latency (hardware) | Policy inference time on Jetson Orin NX (TensorRT)                                      | $\leq 10\text{ ms}$        | Time 1000 forward passes on Jetson Orin NX; report mean $\pm$ std                      | Phase 4 |
| MuJoCo Cross-Validation      | Success rate in MuJoCo as % of Isaac Lab success rate                                   | $\geq 50\%$                | 100 eval episodes in MuJoCo; compare to Isaac Lab baseline                             | Phase 3 |

## 9. Ablation Study Design

All ablations run for 200M steps with 5 independent seeds. Results reported as mean  $\pm$  std of task success rate at convergence.

| ID | Name                | What is varied                                                | Variants                                                   | Expected finding                                                                                    | Seeds         |
|----|---------------------|---------------------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------|
| A1 | Curiosity mechanism | Which intrinsic reward module is used                         | RND / DRND / RDD / ICM / None (PPO only)                   | DRND or RDD > RND > ICM >> None; confirms intrinsic motivation is necessary                         | 5 per variant |
| A2 | Critic design       | Whether the critic receives privileged information            | Asymmetric (privileged) / Symmetric (no privilege)         | Asymmetric critic accelerates learning by 30–50% in wall steps                                      | 5 per variant |
| A3 | Visual input        | Whether the actor receives the 64×64 egocentric depth image   | Depth image + proprioception / Proprioception only         | Depth image helps on tasks requiring spatial reasoning (Occluded Retrieval)                         | 5 per variant |
| A4 | Curriculum staging  | Whether tasks are introduced in difficulty order              | Staged curriculum / All tasks simultaneously               | Curriculum improves emergence rate on hard tasks; all-simultaneous leads to catastrophic forgetting | 5 per variant |
| A5 | $\alpha$ annealing  | Whether curiosity weight is annealed                          | Annealed 1.0→0.1 / Fixed $\alpha=1.0$ / Fixed $\alpha=0.1$ | Annealing achieves best of both: exploration early, exploitation later                              | 3 per variant |
| A6 | QFLEX comparison    | Value-guided flow exploration (Wei et al. 2026) vs. RND-based | RDD / QFLEX-adapted for continuous humanoid                | QFLEX may outperform on high-DOF exploration; important baseline for paper                          | 5 per variant |

## 10. Risk Register & Mitigation Strategies

| Risk                                                        | Likelihood | Impact | Mitigation Strategy                                                                                                                                                                           | Fallback                                                                                                                                                     |
|-------------------------------------------------------------|------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tool use does not emerge within compute budget (500M steps) | Medium     | High   | Start with easiest task (Distant Target). Use curriculum learning. Monitor every 50M steps; if no contact events by 100M steps, switch to hybrid approach (light reward shaping + curiosity). | Hybrid: add small contact bonus (+0.01) without specifying which object. Characterise as 'humanoid hard-exploration benchmark' — still publishable.          |
| Sim-to-real gap causes behaviour collapse on physical G1    | Medium     | Medium | Extensive domain randomisation (Section 6.1). Residual RL with hardware constraints. Sim-to-sim validation in MuJoCo before hardware.                                                         | Use ASAP-style delta-action model (fine-tune on 10 real-world rollouts). Even simulation results alone are publishable contribution.                         |
| Dex3-1 grasp limitations prevent reliable tool manipulation | Low        | High   | Design tool objects with cylindrical/prismatic graspable handles matching Dex3-1 geometry. Simplify grasping primitives. Use magnetic or velcro-assisted grasps as a fallback.                | If Dex3-1 is unreliable: use simpler 2-finger gripper approximation in sim; demonstrate on at least stick-pushing tasks that do not require precision grasp. |
| Hardware damage during sim-to-real deployment               | Low        | High   | Safety gantry mandatory for all real-world tests. Torque limits enforced in residual RL. Graduated deployment: static → slow-motion → full-speed. E-stop remote always within reach.          | All real-world experiments halted; paper based on simulation results only. Simulation contribution is still novel and publishable.                           |
| Compute budget exceeded (cluster queue delays / preemption) | Medium     | Medium | Implement SLURM auto-requeue. Run shorter ablations first; full runs in background. Budget ≥600 GPU-hours to have headroom.                                                                   | Reduce number of seeds from 10 to 5 for emergence rate estimate. Focus ablations on A1 (curiosity mechanism) and A2 (critic design) as highest-priority.     |
| Competing paper scoops the exact result before submission   | Medium     | High   | Move quickly. Monitor arXiv weekly (especially Abbeel/Pathak/NVIDIA groups). Frame contribution around the G1 hardware demo — almost no curiosity papers show real-robot results.             | If scooped in simulation: emphasise the real-robot demo as unique contribution. If scooped entirely: shift to negative result / benchmark paper.             |
| Isaac Lab API changes break environment implementation      | Low        | Medium | Pin Isaac Lab version in Docker image from Day 1. Do not update Isaac Lab mid-project unless a critical bug requires it.                                                                      | Roll back to pinned version. Fix environment if API change is unavoidable.                                                                                   |
| Unitree G1 not available (still to be procured)             | Low        | High   | Initiate procurement immediately (pre-phase). Design all simulation work to be hardware-independent. Phases 1–3 are entirely simulation-based.                                                | If G1 arrives after Week 12: compress Phase 4 to 1 week of evaluation. If G1 never arrives: simulation-only paper.                                           |

## 11. Required Resources Checklist

### 11.1 Compute Resources

| Resource                    | Specification                               | Quantity       | When Needed | Source                               |
|-----------------------------|---------------------------------------------|----------------|-------------|--------------------------------------|
| GPU cluster nodes           | ≥4 NVIDIA A100 80GB or H100 80GB            | ≥4 nodes       | Weeks 1–11  | Purdue RCAC (Anvil/Gilbreth/Negishi) |
| Total GPU-hours             | ~200–500 GPU-hours (full suite + ablations) | 1 allocation   | Weeks 4–11  | RCAC allocation request              |
| Local GPU workstation       | ≥1 RTX 3090 or A5000 per researcher         | 2 workstations | All weeks   | Lab hardware                         |
| Jetson Orin NX (onboard G1) | 275 TOPS, part of G1 EDU                    | 1 (built-in)   | Weeks 12–14 | G1 EDU spec                          |
| Storage                     | ≥1 TB for checkpoints + video               | 1 TB           | All weeks   | RCAC scratch storage                 |

### 11.2 Hardware Resources

| Item                  | Spec                               | Qty | Status               | Action Required                                      |
|-----------------------|------------------------------------|-----|----------------------|------------------------------------------------------|
| Unitree G1 EDU        | 43 DOF, Dex3-1, Jetson Orin NX     | 1   | [existing / procure] | Confirm now; procure if needed (lead time ≈8 weeks)  |
| Safety gantry         | Load-rated ≥100 kg, overhead mount | 1   | [check lab]          | Load-test before Phase 4                             |
| Wooden blocks (tool)  | 30×30×30 cm, 1–3 kg each           | 5   | Procure              | Buy or fabricate in workshop; measure mass precisely |
| PVC rods (tool)       | 80–120 cm length, <0.5 kg          | 5   | Procure              | Buy from hardware store; sand ends smooth            |
| Foam wedges (tool)    | Variable height fulcrum blocks     | 3   | Procure              | Cut from high-density foam                           |
| Ethernet switch       | 1 Gbps managed switch              | 1   | [check lab]          | Confirm <1 ms latency with G1 SDK test               |
| E-stop remote         | Wireless, <50 ms latency           | 2   | [check lab]          | Test wireless range in deployment lab                |
| Overhead RGB-D camera | 1080p@30fps for monitoring only    | 1   | [check lab]          | Mount above task area for video documentation        |

### 11.3 Software Resources

| Software         | Version             | Licence         | Purpose                     |
|------------------|---------------------|-----------------|-----------------------------|
| NVIDIA Isaac Lab | 4.x (Isaac Sim 4.x) | Free (academic) | Primary training simulator  |
| unitree_rl_gym   | Latest main         | BSD-3           | G1 URDF + PPO boilerplate   |
| rsl_rl           | 1.0.x               | BSD-3           | PPO implementation          |
| MuJoCo 3.x       | 3.1+                | Apache 2.0      | Sim-to-sim cross-validation |

| Software            | Version                     | Licence                | Purpose                        |
|---------------------|-----------------------------|------------------------|--------------------------------|
| TensorRT            | 8.6+<br>(Jetson-compatible) | NVIDIA<br>proprietary  | Onboard inference optimisation |
| unitree_sdk2_python | Latest                      | Unitree<br>proprietary | Hardware communication         |
| ROS 2 Humble        | Humble Hawksbill            | Apache 2.0             | Hardware state/command bridge  |
| PyTorch             | 2.1+                        | BSD-3                  | Neural network training        |
| Weights & Biases    | Latest                      | Free (academic)        | Experiment tracking            |
| Python 3.10+        | 3.10 or 3.11                | PSF                    | All scripting                  |



## 12. Literature Positioning & Novelty Arguments

This section documents exactly how to position this work against the most dangerous adjacent papers. Every reviewer will ask about these.

| Paper                                                  | How it differs from our work                                                                                                                                                            | Our distinction                                                                                                                          |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| HumanoidBench (Sferrazza/Abbeel, RSS 2024)             | Same robot family (H1), reports that RL including intrinsic-motivation baselines FAILS on hard whole-body manipulation tasks                                                            | We attack the exact open problem. HumanoidBench is our primary motivating evidence.                                                      |
| ASAP (He et al., RSS 2025)                             | Same G1, same Isaac Gym, whole-body, sim-to-real. BUT: motion-tracking objective (not sparse reward), uses retargeted human motion data (not zero demos), no tools, no novel behaviour. | Demonstration-free, emergent, tool-focused. Reuse their delta-action sim-to-real technique.                                              |
| 'Opening the Sim-to-Real Door' (Wang et al., Dec 2025) | Closest 2025 work on humanoid + RL + manipulation sim-to-real. Uses teacher-student with privileged info; task-specific reward; not curiosity-driven.                                   | No privileged teacher, intrinsic motivation only, free-form emergent tool discovery.                                                     |
| QFLEX (Wei et al., Tsinghua 2026)                      | Value-guided flow exploration for high-DOF continuous control. Strongest pure-exploration method for high-DOF. Does NOT claim emergent tool use, not on humanoid hardware.              | Must treat QFLEX as a primary baseline. Include QFLEX-adapted as ablation A6.                                                            |
| DRND (Yang et al., ICML 2024)                          | Strongest 2024 RND successor for high-DOF manipulation. Validated on Atari + Adroit, NOT on humanoid whole-body control.                                                                | DRND is our primary curiosity baseline. We extend it to 43-DOF humanoid loco-manipulation.                                               |
| DemoHLM (Unitree G1, Oct 2025)                         | Same G1 hardware, 10 loco-manipulation tasks, RGB-D. Still requires one demonstration; not curiosity-driven; not emergent.                                                              | Zero demonstrations. Intrinsic curiosity only.                                                                                           |
| GR00T N1.6 / Helix / Gemini 1.5 (2025–2026)            | Foundation model VLAs trained on massive teleoperation datasets. Different paradigm entirely.                                                                                           | Zero pre-training data. Pure exploration from scratch. Better sample-efficient for novel tools not seen during pre-training.             |
| Baker et al. / OpenAI (ICLR 2020)                      | Canonical emergent tool use reference — hide-and-seek agents. Simplified spherical agents with grab/lock actions, NOT a high-DOF humanoid.                                              | This project is the direct successor: can a single-agent, curiosity-only approach reproduce Baker-style emergence on a bipedal humanoid? |

## 13. Week-by-Week Master Schedule

| Week            | Phase                      | Key Tasks                                                                                                                                                                                                                     | Deliverables / Checkpoints                                                                                |
|-----------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Pre (Before W1) | Pre-Phase Setup            | RCAC accounts, GPU allocation, hardware inventory, G1 procurement, software install, GitHub repo setup, wandb project, safety gantry check                                                                                    | All pre-phase checklist items complete. Isaac Lab running on cluster.                                     |
| Week 1          | Phase 1 Env Design         | Isaac Lab install + cluster test. G1 URDF integration + validation (all 43 DOF). Begin Distant Target + Elevated Button environment code.                                                                                     | G1 URDF loads in Isaac Lab. 100-step random rollout: no NaN. Two env classes started.                     |
| Week 2          | Phase 1 Env Design         | Complete all 5 task environments. Implement procedural randomisation module. Write unit tests for all environments. wandb integration test.                                                                                   | All 5 environments pass unit tests. Randomisation confirmed. wandb live.                                  |
| Week 3          | Phase 1 Env Design         | Full cluster test: 4096 parallel envs, 4xA100, 1 hour. Random-agent baseline success rates. Isaac Lab viewer demo. Phase 1 review meeting.                                                                                    | <b>PHASE 1 EXIT:</b> 4096 envs stable for 1 hour. All 5 envs in viewer demo.                              |
| Week 4          | Phase 2 Curiosity Training | Implement RND, DRND, RDD, ICM modules. Implement asymmetric actor-critic. Set up PPO training loop with dual-head critic. Write observation normalisation.                                                                    | All curiosity modules implemented and unit-tested. Training loop runs for 1M steps.                       |
| Week 5          | Phase 2 Curiosity Training | Launch first full training run on Distant Target task (RND, 500M steps). Run hyperparameter sweep ( $\text{lr}$ , $\text{GAE } \lambda$ , $\text{clip } \epsilon$ ) on 100M steps. Monitor wandb.                             | First training run launched. Sweep results for top-3 hyperparameter configs.                              |
| Week 6          | Phase 2 Curiosity Training | Monitor Distant Target training for emergence events. Launch DRND run. Launch RDD run. Check if <code>tool_contact_count</code> is increasing. If no contact by 100M steps: review curriculum, consider light reward shaping. | First tool-contact event logged (target). Learning curves clearly showing curiosity driving exploration.  |
| Week 7          | Phase 2 Curiosity Training | If Distant Target shows $\geq 20\%$ success: unlock Elevated Button in curriculum. Launch ablation A1 (curiosity mechanisms) runs. Launch ablation A2 (critic design) runs.                                                   | Curriculum Stage 2 unlocked. Ablations A1 + A2 running.                                                   |
| Week 8          | Phase 2 Curiosity Training | Continue full training runs. Launch ablation A3 (visual input) + A4 (curriculum staging). If budget allows: launch ablation A6 (QFLEX comparison). Collect video rollouts every 50M steps.                                    | Ablations A3 + A4 running. At least 3 video recordings of tool-use episodes saved.                        |
| Week 9          | Phase 2 Curiosity Training | Collect final training results. Run 10-seed emergence rate experiment. Trajectory clustering analysis. Phase 2 review meeting. Select best checkpoint for Phase 3 distillation.                                               | <b>PHASE 2 EXIT:</b> $\geq 1$ task at $>20\%$ success. Emergence rate computed. Best checkpoint selected. |
| Week 10         | Phase 3 Distillation       | Dagger distillation (20 iterations). Build demonstration dataset from best curiosity policy. Train student policy. Verify inference $<10$ ms. Begin residual RL: freeze base, train residual MLP.                             | <code>student_policy_v1.pt</code> saved. Inference confirmed $<10$ ms.                                    |
| Week 11         | Phase 3 Distillation       | Complete residual RL (50M steps). Domain randomisation applied. MuJoCo cross-validation (100 eval episodes). Compare deployable policy vs. base in MuJoCo. Phase 3 review meeting.                                            | <b>PHASE 3 EXIT:</b> <code>deployable_policy_v1.pt</code> . MuJoCo $\geq 50\%$ of Isaac Lab success rate. |

| Week    | Phase                    | Key Tasks                                                                                                                                                                            | Deliverables / Checkpoints                                                                       |
|---------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Week 12 | Phase 4<br>Sim-to-Real   | TensorRT export to Jetson Orin NX. Validate latency $\leq 10$ ms on hardware. Stage 1 hardware tests: static tool interaction only, tethered. Physical task scene setup.             | TensorRT running on G1. Stage 1 (static) tests complete, no safety events.                       |
| Week 13 | Phase 4<br>Sim-to-Real   | Stage 2: Slow-motion (25% speed) full task tests, tethered. Run 20 consecutive episodes per task. Log safety events. Begin video documentation.                                      | Stage 2 tests complete. Any instabilities identified and addressed.                              |
| Week 14 | Phase 4<br>Sim-to-Real   | Stage 3: Full-speed evaluation (if Stage 2 clear). 50 real-world evaluation episodes. Record successes. High-quality video recording of successful episodes. Phase 4 review meeting. | <b>PHASE 4 EXIT:</b> 50 episodes evaluated. $\geq 1$ task shows real-world success. Video saved. |
| Week 15 | Phase 5<br>Paper Writing | Write Abstract, Introduction, Related Work, Method sections. Generate all figures and plots. Internal draft review.                                                                  | Full first draft circulated to all co-authors and advisor.                                       |
| Week 16 | Phase 5<br>Paper Writing | Revise draft. Write Conclusion + Limitations. Produce supplementary video. Code release preparation. Submit to target venue.                                                         | <b>PROJECT COMPLETE:</b> Paper submitted. Code released on GitHub. Checkpoints archived.         |

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*End of Implementation Plan — Curiosity-Driven RL for Emergent Tool Use on Humanoid Robot*

*Purdue University · School of Mechanical Engineering · April 2026*